Limit:

Note: Use one sided limits for the functions which shows jumps, undefined behavior at point $x = a$ or for piecewise functions.

2. $\lim_{x \to 0} \frac{x}{|x|}$.

$$|x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$$

$$\frac{x}{|x|} = \begin{cases} 1, & x \geq 0 \\ -1, & x < 0 \end{cases}$$

$$L.H.L. = \lim_{x \to 0^-} \frac{x}{|x|} = \lim_{x \to 0^-} (-1)$$

$$= -1$$

$$R.H.L. = \lim_{x \to 0^+} (1) = 1$$

$\therefore L.H.L. \neq R.H.L$

$\therefore$ $\boxed{\text{So, limit doesn't exist}}$

Note: for $x \to \infty$, try to have functions of type $\left(\frac{1}{x}\right)$.

7. $\lim_{x \to \infty} \frac{3x+5}{6x+8} = \lim_{x \to \infty} \frac{3+5/x}{6+8/x} = \frac{3}{6} = \boxed{\frac{1}{2}}$

[We don't have to use One sided limits here]

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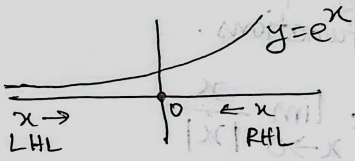
Continuity:

3. Test the continuity of $f(x) = \begin{cases} e^{1/x}, & x \neq 0 \\ 1, & x = 0 \end{cases}$ at $x=0$

Sol^n:

(i) $f(0) = 1$



(ii) $\text{L.H.L.} = \lim_{x \to 0^-} e^{1/x} = 0$

[x is negative, so $\frac{1}{x}$ is also negative, hence $e^{-\infty}$]

$\text{R.H.L.} = \lim_{x \to 0^+} e^{1/x} = \infty$

$\text{LHL} \neq \text{RHL}$

So limit doesn't exist.

$\boxed{\text{Hence } f(x) \text{ is not continuous at } x=0.}$

Differentiability:

Note: We have to check if L.H.D. = R.H.D regardless of what type of function.

1. Check the differentiability at $x = \frac{\pi}{2}$ for

$$f(x) = \begin{cases} 1 + \sin x, & 0 \le x < \pi/2 \\ 2 + \left(x - \frac{\pi}{2}\right)^2, & x \geq \pi/2 \end{cases}$$

Sol^n:

$$\text{L.H.D.} = \lim_{h \to 0^-} \frac{f\left(\frac{\pi}{2} + h\right) - f(\pi/2)}{h}$$

$\boxed{\text{for } h \to 0^- \text{ (h are negatives)}, \ \frac{\pi}{2} + h < \frac{\pi}{2}}$

$$= \lim_{h \to 0^-} \frac{1 + \sin\left(\frac{\pi}{2} + h\right) - 2 - \left(\frac{\pi}{2} - \frac{\pi}{2}\right)^2}{h}$$

$$= \lim_{h \to 0^-} \frac{1 + \cos h - 2}{h} = \lim_{h \to 0^-} \frac{\cos h - 1}{h}$$

$$= \lim_{h \to 0^-} (-\sin h) \ \boxed{\text{L Hospital's rule}} = 0$$

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$$R.H.D. = \lim_{h \to 0^+} \frac{f\left(\frac{\pi}{2}+h\right) - f\left(\frac{\pi}{2}\right)}{h}$$

[for $h \to 0^+$ (h are positive), $\frac{\pi}{2}+h > \frac{\pi}{2}$]

$$= \lim_{h \to 0^+} \frac{2+\left(\frac{\pi}{2}+h-\frac{\pi}{2}\right)^2 - 2 - \left(\frac{\pi}{2}-\frac{\pi}{2}\right)^2}{h}$$

$$= \lim_{h \to 0^+} \frac{2+h^2-2-0}{h}$$

$$= \lim_{h \to 0^+} \frac{h^2}{h} = \lim_{h \to 0^+} h = 0$$

$\therefore$ L.H.D. = R.H.D.

$\therefore$ $f(x)$ is differentiable at $x = \frac{\pi}{2}$.

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